

R A C E D A Y

A P I E N D P O I N T P L A N

API Specification

This document specifies every API endpoint the RaceDay system will expose.
It was completed before any API code was written. The implemented API must closely match this plan.

1. Conventions

Item	Convention
Base URL	/api
Auth mechanism	JWT Bearer Token
Role enforcement	Endpoints marked with a role require a valid JWT with the corresponding role claim
ID parameters	All {id} path parameters are integers
Date format	ISO 8601 (YYYY-MM-DD)
Time format	HH:MM:SS

2. Authentication

2.1 POST /api/auth/register

Route	POST /api/auth/register
Description	Register a new user account. The request must specify the desired role.
Role Required	None (public)
Request Body	{ "firstName": "string", "lastName": "string", "email": "string", "password": "string", "role": "Organiser Participant" }
Responses	• 201 Created user object with token • 400 Bad Request validation errors • 409 Conflict email already registered

2.2 POST /api/auth/login

Route	POST /api/auth/login
Description	Authenticate a user and return a JWT token.
Role Required	None (public)
Request Body	{ "email": "string", "password": "string" }
Responses	• 200 OK { "token": "string", "user": { ... } } • 401 Unauthorized invalid credentials

3. User Profile

3.1 GET /api/profile

Route	GET /api/profile
Description	Retrieve the authenticated user's profile details.
Role Required	Organiser or Participant
Request Body	None
Responses	• 200 OK user profile object • 401 Unauthorized not authenticated

3.2 PUT /api/profile

Route	PUT /api/profile
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Description	Update the authenticated user's profile (name, phone, city, province, etc.). Password and email cannot be changed here.
Role Required	Organiser or Participant
Request Body	{ "firstName": "string", "lastName": "string", "dateOfBirth": "date", "gender": "string", "phoneNumber": "string", "city": "string", "province": "string" }
Responses	• 200 OK updated profile object • 400 Bad Request validation errors • 401 Unauthorized not authenticated

4. Events

4.1 GET /api/events

Route	GET /api/events
Description	Retrieve all published events. Supports optional query parameters for filtering by eventType, city, province, and status.
Role Required	None (public)
Request Body	None
Responses	• 200 OK array of event summary objects

4.2 GET /api/events/{id}

Route	GET /api/events/{id}
Description	Retrieve full details of a specific event, including its categories.
Role Required	None (public)
Request Body	None
Responses	• 200 OK event object with categories • 404 Not Found event does not exist

4.3 POST /api/events

Route	POST /api/events
Description	Create a new event. The OrganiserId is set from the authenticated user's token.
Role Required	Organiser
Request Body	{ "name": "string", "description": "string", "eventDate": "date", "startTime": "time", "location": "string", "city": "string", "province": "string", "eventType": "Running Walking Cycling" }
Responses	• 201 Created created event object • 400 Bad Request validation errors • 401 Unauthorized not authenticated • 403 Forbidden not an Organiser

4.4 PUT /api/events/{id}

Route	PUT /api/events/{id}
Description	Update an existing event. Only the organiser who created the event can update it.
Role Required	Organiser
Request Body	{ "name": "string", "description": "string", "eventDate": "date", "startTime": "time", "location": "string", "city": "string", "province": "string", "eventType": "string", "status": "string" }

Responses	• 200 OK updated event object • 400 Bad Request validation errors • 403 Forbidden not the event organiser • 404 Not Found event does not exist
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4.5 DELETE /api/events/{id}

Route	DELETE /api/events/{id}
Description	Delete an event. Only the organiser who created it can delete it. Cascades to categories, enrolments, and results.
Role Required	Organiser
Request Body	None
Responses	• 204 No Content successfully deleted • 403 Forbidden not the event organiser • 404 Not Found event does not exist

4.6 GET /api/events/my-events

Route	GET /api/events/my-events
Description	Retrieve all events created by the authenticated organiser.
Role Required	Organiser
Request Body	None
Responses	• 200 OK array of event objects

5. Categories

5.1 GET /api/events/{eventId}/categories

Route	GET /api/events/{eventId}/categories
Description	Retrieve all categories for a specific event.
Role Required	None (public)
Request Body	None
Responses	• 200 OK array of category objects • 404 Not Found event does not exist

5.2 GET /api/categories/{id}

Route	GET /api/categories/{id}
Description	Retrieve a specific category by its ID.
Role Required	None (public)
Request Body	None
Responses	• 200 OK category object • 404 Not Found category does not exist

5.3 POST /api/events/{eventId}/categories

Route	POST /api/events/{eventId}/categories
Description	Create a new category under a specific event. Only the organiser of that event can create categories.
Role Required	Organiser
Request Body	{ "name": "string", "distance": 0.00, "maxParticipants": 0, "entryFee": 0.00 }
Responses	• 201 Created created category object • 400 Bad Request validation errors • 403 Forbidden not the event organiser • 404 Not Found event does not exist

5.4 PUT /api/categories/{id}

Route	PUT /api/categories/{id}
Description	Update an existing category. Only the organiser of the parent event can update it.
Role Required	Organiser
Request Body	{ "name": "string", "distance": 0.00, "maxParticipants": 0, "entryFee": 0.00 }
Responses	• 200 OK updated category object • 403 Forbidden not the event organiser • 404 Not Found category does not exist

5.5 DELETE /api/categories/{id}

Route	DELETE /api/categories/{id}
Description	Delete a category. Only the organiser of the parent event can delete it.
Role Required	Organiser
Request Body	None
Responses	• 204 No Content successfully deleted • 403 Forbidden not the event organiser • 404 Not Found category does not exist

6. Enrolments

6.1 POST /api/enrolments

Route	POST /api/enrolments
Description	Enrol the authenticated participant into a specific event category. The ParticipantId is set from the token.
Role Required	Participant
Request Body	{ "categoryId": 0 }
Responses	• 201 Created enrolment object • 400 Bad Request validation errors • 404 Not Found category does not exist • 409 Conflict already enrolled in this category

6.2 GET /api/enrolments/my-enrolments

Route	GET /api/enrolments/my-enrolments
Description	Retrieve all enrolments for the authenticated participant, including event and category details.
Role Required	Participant
Request Body	None
Responses	• 200 OK array of enrolment objects with event/category info

6.3 GET /api/events/{eventId}/enrolments

Route	GET /api/events/{eventId}/enrolments
Description	Retrieve all enrolments for a specific event. Only the organiser of that event can view this.
Role Required	Organiser
Request Body	None
Responses	• 200 OK array of enrolment objects with participant info • 403 Forbidden not the event organiser • 404 Not Found event does not exist

6.4 DELETE /api/enrolments/{id}

Route	DELETE /api/enrolments/{id}
Description	Withdraw from an event (delete the enrolment). Only the enrolled participant can withdraw.
Role Required	Participant
Request Body	None
Responses	• 204 No Content successfully withdrawn • 403 Forbidden not the enrolled participant • 404 Not Found enrolment does not exist

6.5 PUT /api/enrolments/{id}/status

Route	PUT /api/enrolments/{id}/status
Description	Update the status of an enrolment (e.g., Confirmed, DNS). The organiser of the parent event can update status.
Role Required	Organiser
Request Body	{ "status": "Registered Confirmed Withdrawn DNS" }
Responses	• 200 OK updated enrolment object • 400 Bad Request invalid status value • 403 Forbidden not the event organiser • 404 Not Found enrolment does not exist

7. Results

7.1 POST /api/results

Route	POST /api/results
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Description	Capture a result for a specific enrolment. Only the organiser of the parent event can capture results.
Role Required	Organiser
Request Body	{ "enrolmentId": 0, "finishTime": "HH:MM:SS", "position": 0, "status": "Finished DNF DSQ" }
Responses	• 201 Created result object • 400 Bad Request validation errors • 404 Not Found enrolment does not exist • 409 Conflict result already recorded for this enrolment

7.2 GET /api/results/{id}

Route	GET /api/results/{id}
Description	Retrieve a specific result by its ID.
Role Required	Organiser or Participant
Request Body	None
Responses	• 200 OK result object • 404 Not Found result does not exist

7.3 GET /api/events/{eventId}/results

Route	GET /api/events/{eventId}/results
Description	Retrieve all results for a specific event, grouped by category.
Role Required	None (public)
Request Body	None
Responses	• 200 OK array of result objects with participant and category info • 404 Not Found event does not exist

7.4 GET /api/results/my-results

Route	GET /api/results/my-results
Description	Retrieve all results for the authenticated participant across all events.
Role Required	Participant
Request Body	None
Responses	• 200 OK array of result objects with event and category info

7.5 PUT /api/results/{id}

Route	PUT /api/results/{id}
Description	Update an existing result. Only the organiser of the parent event can update.
Role Required	Organiser
Request Body	{ "finishTime": "HH:MM:SS", "position": 0, "status": "Finished DNF DSQ" }
Responses	• 200 OK updated result object • 403 Forbidden not the event organiser • 404 Not Found result does not exist

7.6 DELETE /api/results/{id}

Route	DELETE /api/results/{id}
Description	Delete a result record. Only the organiser of the parent event can delete.
Role Required	Organiser
Request Body	None
Responses	• 204 No Content successfully deleted • 403 Forbidden not the event organiser • 404 Not Found result does not exist

8. Endpoint Summary

Resource Area	Count
Authentication	2
User Profile	2
Events	6
Categories	5
Enrolments	5
Results	6
Total	26

9. Error Response Format

All error responses follow a consistent JSON structure, per the RFC 7807 Problem Details standard used by ASP.NET Core:

```
{
  "type": "string",
  "title": "string",
  "status": 0,
  "detail": "string",
  "errors": {
    "fieldName": ["error message"]
  }
}
```